

7. Ethics and Ownership

7.1. Ethics and Ownership

- Ethics: a system of moral principles that guide behaviour based on philosophical views

- Computer Ethics
 - Regulate how computing professionals should make decisions regarding professional & social conduct.
 - A computing professional can be ethically guided by joining a professional, ethical body such as the BCS and IEEE, which have codes of conduct.
 - The BCS Code of Conduct provides the following guidance:
 - **Public Interest:** Both codes emphasize the importance of acting in the public interest and safeguarding the health, safety, and welfare of the public.
 - **Judgement and Integrity:** Professionals are expected to exercise sound judgement and maintain their integrity in their work.
 - **Ethical Practice:** Both codes stress the importance of ethical behaviour and decision-making.
 - **Lifelong Learning:** The IEEE code explicitly mentions the need for continuous learning, which aligns with the need to maintain professional competence.
 - The IEEE Software Engineering Code of Ethics defines eight principles:
 - **PUBLIC:** Software engineers shall act consistently with the public interest.
 - **CLIENT AND EMPLOYER:** Software engineers shall act in the best interests of their client and employer, while also considering the public interest.
 - **PRODUCT:** Software engineers shall ensure that their products meet the highest professional standards possible.
 - **JUDGEMENT:** Software engineers shall maintain integrity and independence in their professional judgement.
 - **MANAGEMENT:** Software engineering managers and leaders shall promote an ethical approach to the management of software development and maintenance.
 - **PROFESSION:** Software engineers shall advance the integrity and reputation of the profession, consistent with the public interest.
 - **COLLEAGUES:** Software engineers shall be fair to and supportive of their colleagues.
 - **SELF:** Software engineers shall participate in lifelong learning and promote an ethical approach to their practice.

- Ownership
 - Data ownership: having legal rights and complete control over a single piece or set of data elements.
 - Copyright gives the creators of some types of media rights to control how they're used and distributed.
 - Competitors can steal programming ideas and methods; software can easily be copied and bootlegged (sold illegally); hence, legislation is needed to protect the ownership, usage, and copyright of data.

Software Licencing

- Free Software Foundation:
 - A license gives users freedom to run, copy, distribute, study, change, and improve the software.
 - Condition: any redistributed version of software must be distributed with original terms of free use, modification, and distribution (aka copyleft)
- The Open Source Initiative:
 - The source code of open-source software is readily available to users under copyright; it does enable users to re-distribute the software.
 - The concept of an open-source program relies on the fact that users can review source code to eliminate bugs in it
- Shareware:
 - Demonstration software that is distributed for free but for a specific evaluation period only
 - Distributed on a trial basis and with an understanding that sometime later, a user may be interested in paying for it
 - Used for marketing purposes
- Commercial: Requires payment before it can be used, but includes all program's features, with no restrictions

Artificial intelligence (AI)

- Artificial Intelligence (AI): the ability of a computer to perform tasks in such a way that is conventionally associated with human intelligence:
 - AI can learn from past mistakes
 - they adapt to stop the same problem from occurring again
 - they learn to predict what might happen & raise alert
 - AI can learn to work more efficiently
 - when an action slows the system down, it can prevent this from happening again
 - when an action increases the speed of the system, it can repeat this when necessary to improve efficiency

- AI Applications
 - Developing autonomous mechanical products
 - Machine learning through data sets
- AI Impacts
 - Social
 - Replacement of manual labour with automation could lead to massive unemployment.
 - However, it could lead to increased leisure time.
 - Economic: Due to increased innovation and efficiency with automation provided by AI, there'd be lower manufacturing costs in general
 - Environmental: Detrimental impact on the environment due to robot manufacture with limited resources and its waste disposal